JAM SADIQ

ORCID: 0000-0001-5931-3624 ◇ (+34) 604 971 805 ◇ jsadiq@ifae.es ◇ jam-sadiq.github.io
IFAE – Institut de Física d'Altes Energies, Barcelona, Spain

EMPLOYMENT AND EDUCATION

Research Software Engineer *November 2025 – present*
IFAE – Institut de Física d'Altes Energies & PIC – Port d'Informació Científica,
Universitat Autònoma de Barcelona, Spain
Supporting IGWN computing infrastructure and research for the LIGO–Virgo–KAGRA collaboration: HTCondor-based Linux cluster systems, containerisation of scientific workflows (Docker/Singularity), gravitational-wave data analysis.

Data Engineer *May 2025 – October 2025*
VitalizeDx, Trieste, Italy
Developed data pipelines and applied deep-learning techniques to medical data analysis.

Postdoctoral Researcher *February 2023 – February 2025*
SISSA (International School for Advanced Studies), Trieste, Italy

Postdoctoral Researcher *October 2020 – September 2022*
Instituto Galego de Física de Altas Enerxías (IGFAE), University of Santiago de Compostela, Spain

Assistant Professor in Mathematics *September 2019 – August 2020*
Department of Mathematics and Social Sciences, Sukkur IBA University, Sukkur, Sindh, Pakistan
Courses: General Relativity, Quantum Mechanics, Computational Physics, Introductory Machine Learning

PhD in Astrophysics (Fulbright PhD Fellow) *August 2014 – August 2019*
Center for Computational Relativity and Gravitation, Rochester Institute of Technology, Rochester, NY, USA

TECHNICAL SKILLS

GW software	PyCBC, GWpy, LALSuite, RIFT, Bilby, LISA Analysis Tools
ML / numerics	scikit-learn, PyTorch, JAX, Mathematica, SageMath
DevOps	Git, GitLab CI/CD, Bitbucket, Dagger
Infrastructure	HTCondor, Docker, Singularity (job containerisation)
Languages	Python, C/C++, R, Bash
Operating systems	Linux (Fedora, CentOS, RHEL, Ubuntu), macOS, Windows

GRANTS AND FELLOWSHIPS

ACME Transnational Access Grant *September 2026*
Competitive research-visit grant to the L2IT Centre of Expertise (Toulouse, France)
EU Horizon Europe project ACME, Grant Agreement No. 101131928

Postdoctoral Fellowship *October 2020 – September 2022*
Xunta de Galicia postdoctoral research fellowship at IGFAE, University of Santiago de Compostela, Spain

Fulbright PhD Fellowship *August 2014 – August 2019*
Five-year Fulbright PhD fellowship, U.S. Department of State

Japanese need based Fellowship *September 2010 – January 2012*
Merit based MPhil fellowship at Quaid-i-azam University, Islamabad, Pakistan

RESEARCH PROJECTS

Spinning Subpopulations of Binary Black Holes: From GWTC-3 to GWTC-5
Using an iterative KDE-based reconstruction of the LVK binary black hole (BBH) population, we confirmed a low, positive effective spin for low-mass mergers and identified a potential high-spin subpopulation ($\chi_{\rm eff} \sim 0.45$) at primary masses above $35\,M_\odot$, suggesting a distinct formation channel (Phys. Rev. D). I am now extending this analysis to search for spinning subpopulations at low masses using GWTC-5 BBH observations (in preparation).

Anti-Glitch Search in the PyCBC Pipeline
Developing an anti-glitch search integrated into the PyCBC pipeline that follows the same framework as the standard CBC search but uses analytical glitch-model waveforms, enabling identification of several glitch types and improving the sensitivity and robustness of PyCBC searches (ongoing).

K-dynamics: Well-Posed 1+1 Evolutions in K-essence Using an Implicit Solver
Developing an implicit finite-difference scheme to evolve scalar fields in spherically symmetric spacetimes, aiming to demonstrate the well-posedness of specific K-essence models – modifications of general relativity relevant to inflation and dark energy (in preparation).

Reconstructing the Massive Black Hole Binary Population with LISA via Iterative KDE
Investigated massive black hole binaries (MBHBs) detectable by LISA, focusing on Population III remnants. Proposed an iterative, non-parametric KDE method that accounts for selection effects and addresses statistical challenges across mass and redshift, to reconstruct the astrophysical MBHB distribution from limited observations (Phys. Rev. D).

Non-Parametric Population Inference of Binary Black Holes via Iterative Adaptive KDE
Developed a fast, self-consistent, iteratively reweighted adaptive-bandwidth KDE framework, accounting for selection effects, to reconstruct BBH populations directly from GW observations. Applications include one- and two-dimensional mass distributions and their correlations, tests for redshift evolution of the mass spectrum in GWTC-3, and a method for detecting peaks in a distribution and testing their significance (three papers in ApJ and Phys. Rev. D, 2022–2025).

Waveform Modelling and Numerical Relativity Comparisons
Constructed high-accuracy hybrid waveforms for precessing BBHs by matching post-Newtonian inspiral to numerical-relativity merger waveforms, enabling precise parameter estimation for current and future GW detections (2020). Earlier, introduced a gauge-independent technique based on geodesic deviation to assess analytical spacetime models against numerical relativity in BBH systems (2018).

LIGO Collaboration & LISA Pipeline Development
As an LSC member, automated and enhanced the PyGRB search pipeline. Also developed a simulation-based inference (SBI) detection pipeline tailored to LISA observations.

CONFERENCES AND PRESENTATIONS

Throughput Computing Week 2026, Madison, Wisconsin, USA, June 8–11, 2026 (attended)

LIGO–Virgo–KAGRA Collaboration Meeting, Pisa, Italy, March 9–13, 2026 (attended)

Invited talk, Institute for Cosmic Ray Research (ICRR), University of Tokyo, Japan, January 26, 2026
Black hole binary subpopulations revealed by model-free reconstruction

34th Workshop on General Relativity and Gravitation in Japan (JGRG34), YITP, Kyoto, January 19–23, 2026
Talk: Probing Evolution in the Black Hole Spectrum With Gravitational-Wave Catalogs

Invited talk, Strong Gravity Group Seminar, Niels Bohr Institute, Copenhagen, Denmark, January 31, 2025
Looking To The Horizon: Hints of an Evolving Black Hole Spectrum with GWTC-3?

LISA Spain Meeting, October 15–16, 2024
Contributed talk: Population Analysis of PopIII Massive Black Hole Binaries with LISA using Iterative Weighted KDE

First TEONGRAV International Workshop on the Theory of Gravitational Waves, Sapienza Rome, September 16–20, 2024

Contributed talk: Population Properties of Massive Binary Black Holes with LISA Observations Using Iterative Reweighted KDE

Amaldi15 Conference on Gravitational Waves (online), July 17–21, 2023
Contributed talk: Binary vision: The merging black hole binary mass distribution via iterative density estimation

LVK Meeting, Cardiff, UK, September 2022
Poster: Flexible and Fast Estimation of Binary Merger Population Distributions with Adaptive KDE

GWPAW 2021, Hannover, Germany, December 14–17, 2021
Poster: Flexible and Fast Estimation of Binary Merger Population Distributions with Adaptive KDE

Amaldi14 Conference on Gravitational Waves (online), July 19–23, 2021
Poster: Flexible and Fast Estimation of Binary Merger Population Distributions with Adaptive KDE

11th Iberian Gravitational Waves Meeting (online), June 9–11, 2021
Talk: Hybrid Waveforms for Precessing Binary Black Holes for LIGO Data Analysis

APS April Meeting, Denver, Colorado, USA, April 13–16, 2019
Talk: Hybrid Waveforms for Precessing Binary Black Holes for LIGO Data Analysis

APS April Meeting, Columbus, Ohio, USA, April 14–17, 2018
Talk: A New Tool to Check the Accuracy of Analytic Spacetimes for Binary Black Holes

27th Midwest Relativity Meeting, Ann Arbor, MI, USA, October 12–14, 2017
Talk: Comparing Spacetimes using Geometric Scalars

21st International Conference on General Relativity and Gravitation (GR21), Columbia University, New York, USA, July 10–15, 2016
Talk: Comparing Spacetimes using Geometric Scalars

WORKSHOPS AND TRAINING

13th LISA Cosmology Working Group Workshop, June 1–5, 2026
Institute of Cosmos Sciences (ICCUB), University of Barcelona, Spain

BSC Training Course: Research Data Management for Beginners, April 27–29, 2026
Barcelona Supercomputing Center (BSC), Barcelona, Spain

First AI-INFN Advanced Hackathon, November 26–28, 2024
University of Padua, Italy

Emerging Methods in GW Population Inference, June 24–27, 2024
Institute for Fundamental Physics of the Universe (IFPU), Trieste, Italy

10th Workshop on Collaborative Scientific Software Development and Management of Open Source Scientific Packages, November 6–17, 2023
ICTP, Trieste, Italy

European Einstein Toolkit School 2022, August 29 – September 2, 2022
University College Dublin, Ireland

Current Challenges in Gravitational Physics Workshop (online), April 21–28, 2021
SISSA, Italy

Computational Challenges in Gravitational Wave Astronomy, January 28 – February 1, 2019
IPAM, University of California, Los Angeles, USA

North American Einstein Toolkit School and Workshop, July 31 – August 4, 2017
NCSA, University of Illinois at Urbana-Champaign, USA

Astrophysical Black Hole Merger Workshop VI, June 21–22, 2016 (organising committee member)
Center for Computational Relativity and Gravitation, RIT, Rochester, NY, USA

Caltech Gravitational Wave Astrophysics School 2015, July 6–10, 2015
California Institute of Technology, USA

MENTORING AND TEACHING

Supervised master's thesis: Zainab Mangi (Sukkur IBA University), *Improved hybrid waveforms for generic precessing binary black holes*, 2023. Now a Fulbright PhD candidate at CCRG, RIT, USA.

Supervised undergraduate final-year project: Fahad Ahmed (Sukkur IBA University), *Dynamics of binary systems in Newtonian gravity and general relativity*, 2024. Now a master's student at Université Marie et Louis Pasteur, Besançon, France.

Co-advised summer undergraduate students: Ifra Hassan and Nilo Cher, RIT, on visualising two- and three-body Newtonian dynamics, 2018.

Taught general relativity, quantum mechanics and computational physics to advanced undergraduates, Sukkur IBA University, 2019–2020.

Taught differential equations and linear algebra to advanced undergraduates, UET Taxila (Chakwal sub-campus), 2012–2013.

PUBLIC OUTREACH

SISSA for Schools (S4S) volunteer, Trieste, Italy, 2023–2024.

Public lecture on astronomy and Stephen Hawking's contributions to cosmology, Sukkur IBA University, Pakistan, 2019.

Volunteer, World Science Festival and **Science Ambassador, City of Science**, New York, USA, 2017–2018.

Member, Astronomy Section of the Rochester Academy of Science (ASRAS); public outreach events, 2016–2018.

REFERENCES

Yosef Zlochower (PhD advisor)
Rochester Institute of Technology
yrzsma@rit.edu

Andrew Lundgren
ICREA & IFAE, Barcelona
alundgren@ifae.es

Thomas Dent
University of Santiago de Compostela
thomas.dent@usc.es

Enrico Barausse (Associate Professor)
SISSA, Trieste, Italy
barausse@sissa.it

PUBLICATIONS

A. Selected first- and senior-author publications

1. A three-dimensional atlas of the LIGO–Virgo–KAGRA binary black hole population
   Thomas Dent, **Jam Sadiq**, Ana Lorenzo-Medina, Juan Calderón Bustillo, *in preparation*

2. Seeking spinning subpopulations of black hole binaries via iterative density estimation
   Jam Sadiq, Thomas Dent, Ana Lorenzo-Medina, **Phys. Rev. D** (accepted, 2025)
   arXiv:2506.02250

3. Looking to the horizon: probing evolution in the black hole spectrum with GW catalogs
   Jam Sadiq, Thomas Dent, Ana Lorenzo-Medina, **Phys. Rev. D** (2025)
   arXiv:2502.06451

4. Reconstructing the LISA massive black hole binary population via iterative kernel density estimation
   Jam Sadiq, Kallol Dey, Thomas Dent, Enrico Barausse, **Phys. Rev. D 111, 063051** (2025)
   arXiv:2410.17056

5. Binary vision: the merging black hole binary mass distribution via iterative density estimation
   Jam Sadiq, Thomas Dent, Mark Gieles, **Astrophys. J. 960, 65** (2024)
   doi:10.3847/1538-4357/ad0ce6

6. Flexible and fast estimation of binary merger population distributions with adaptive KDE
   Jam Sadiq, Thomas Dent, Daniel Wysocki, **Phys. Rev. D 105, 123014** (2022)
   doi:10.1103/PhysRevD.105.123014, arXiv:2112.12659

7. Hybrid waveforms for generic precessing binaries for gravitational-wave data analysis
   Jam Sadiq, Yosef Zlochower, Richard O'Shaughnessy, Jacob Lange, **Phys. Rev. D 102, 024012** (2020)
   doi:10.1103/PhysRevD.102.024012, arXiv:2001.07109

8. Comparing an analytical spacetime metric for a merging binary to a fully nonlinear numerical evolution using curvature scalars
   Jam Sadiq, Yosef Zlochower, Hiroyuki Nakano, **Phys. Rev. D 97, 084007** (2018)
   doi:10.1103/PhysRevD.97.084007, arXiv:1802.02990

B. Other contributed publications

1. General-relativistic precession in a black-hole binary
   M. Hannam, C. Hoy, J. E. Thompson et al. (incl. **Jam Sadiq**), **Nature 610, 652–655** (2022)
   doi:10.1038/s41586-022-05212-z

C. Selected LIGO–Virgo–KAGRA collaboration publications

Full list available via ORCID.

1. GWTC-4.0: Population properties of merging compact binaries
   LIGO Scientific, Virgo and KAGRA Collaborations, arXiv:2508.18083 (2025)

2. Observation of gravitational waves from the coalescence of a 2.5–4.5 $M_\odot$ compact object and a neutron star
   LIGO Scientific, Virgo and KAGRA Collaborations, **Astrophys. J. Lett. 970, L34** (2024)
   doi:10.3847/2041-8213/ad5beb, arXiv:2404.04248

3. A joint Fermi-GBM and Swift-BAT analysis of gravitational-wave candidates from the third observing run
   LIGO Scientific, Virgo and KAGRA Collaborations and Fermi GBM Team, **Astrophys. J. 964, 149** (2024)
   doi:10.3847/1538-4357/ad1eed, arXiv:2308.13666

4. Population of merging compact binaries inferred using gravitational waves through GWTC-3
   LIGO Scientific, Virgo and KAGRA Collaborations, **Phys. Rev. X 13, 011048** (2023)
   doi:10.1103/PhysRevX.13.011048

5. Search for gravitational waves associated with gamma-ray bursts detected by Fermi and Swift during the LIGO–Virgo run O3b
LIGO Scientific, Virgo and KAGRA Collaborations, **Astrophys. J. 928, 186** (2022)
doi:10.3847/1538-4357/ac532b, arXiv:2111.03608

6. Search for intermediate-mass black hole binaries in the third observing run of Advanced LIGO and Advanced Virgo
LIGO Scientific, Virgo and KAGRA Collaborations, **Astron. Astrophys. 659, A84** (2022)
doi:10.1051/0004-6361/202141452

7. GWTC-2: Compact binary coalescences observed by LIGO and Virgo during the first half of the third observing run
LIGO Scientific and Virgo Collaborations, **Phys. Rev. X 11, 021053** (2021)
doi:10.1103/PhysRevX.11.021053, arXiv:2010.14527

8. All-sky search in early O3 LIGO data for continuous gravitational-wave signals from unknown neutron stars in binary systems
LIGO Scientific and Virgo Collaborations, **Phys. Rev. D 103, 064017** (2021)
doi:10.1103/PhysRevD.103.064017, arXiv:2012.12128

9. Search for gravitational waves associated with gamma-ray bursts detected by Fermi and Swift during the LIGO–Virgo run O3a
LIGO Scientific and Virgo Collaborations, **Astrophys. J. 915, 86** (2021)
arXiv:2010.14550

10. Gravitational-wave constraints on the equatorial ellipticity of millisecond pulsars
LIGO Scientific and Virgo Collaborations, **Astrophys. J. Lett. 902, L21** (2020)
doi:10.3847/2041-8213/abb655, arXiv:2007.14251